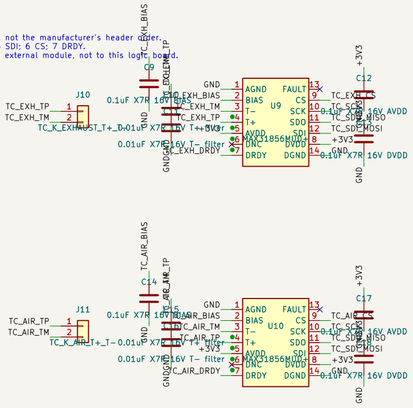
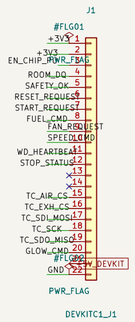
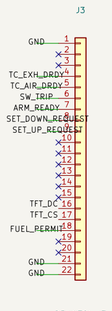
SIROKKO 262.01 — COMMON CONTROLLER INTERFACES / DRAFT
ESP32-S3-DevKitC-1-N8R8 v1.1: J1 and J3 numbers follow the Espressif Header Block.
Native KiCad project. This sheet is NOT the complete heater wiring or a manufacturing release.

EXTERNAL SENSOR MODULES
J10/J11 pin order is OUR HARNESS, not the manufacturer's header order.
1 Vin=3V3; 2 GND; 3 SCK; 4 SDO; 5 SDI; 6 CS; 7 DRDY.
K probes connect to T+/T− of each external module, not to this logic board.





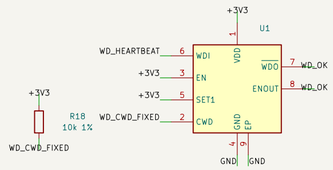


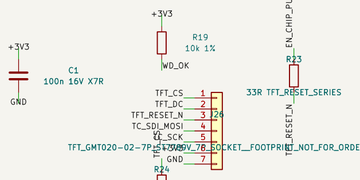
EXTERNAL POWER AND SAFETY STAGES
J20/J21 carry 3V3 LOGIC ONLY. NO 24V, coil or motor current here.
FUEL_PERMIT must pass through independent watchdog + latched limit chain.
STOP must also break the external fuel permission directly.
Protective fuel/glow disconnects and fan off-delay are not implemented on this sheet.

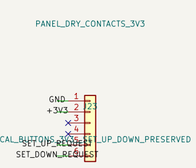


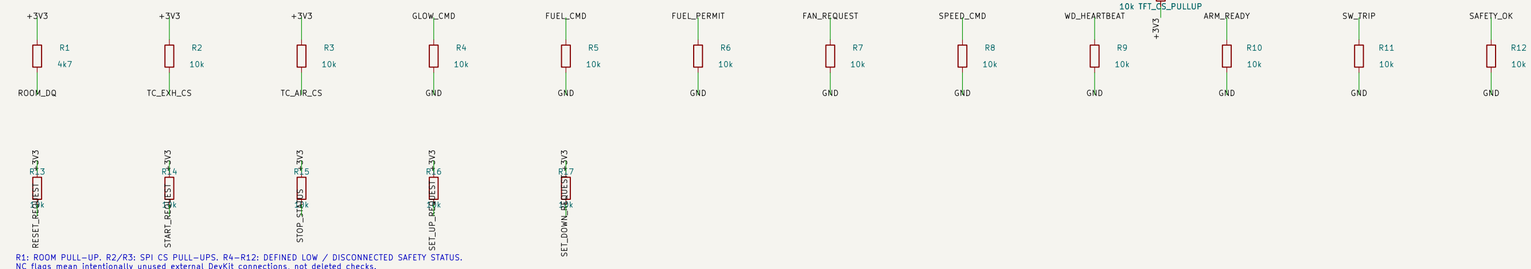
POWER / ROOM SENSOR
J13: externally regulated +5V only, never raw 24V.
Do not combine USB and external supply without verified power isolation.
J12 harness: 1 VDD=3V3; 2 DQ; 3 GND. Actual DS18B20 package pinout differs.





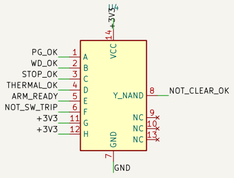


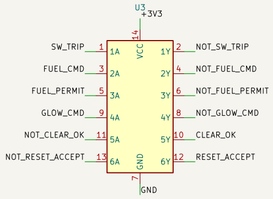


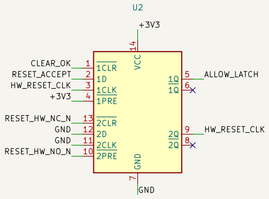


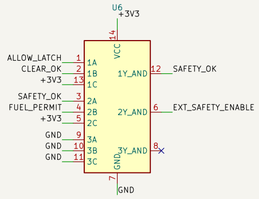
R1: ROOM PULL-UP. R2/R3: SPI CS PULL-UPS. R4–R12: DEFINED LOW / DISCONNECTED SAFETY STATUS.
NC flags mean intentionally unused external DevKit connections, not deleted checks.
Rev A: SPEED_CMD selects low speed through a non-overlapping changeover contact.
Rev B: SPEED_CMD is PWM: low-speed motor lead is isolated; independent full-speed purge path required.
ERC checks this interface network only; it cannot verify combustion, external modules or protective timing.

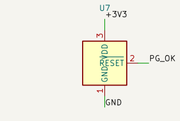
HARDWARE FUEL PERMISSION — DRAFT / NO FIRED OPERATION
U2.1: fault-dominant allow latch. U2.2: SPDT RESET debounce.
RESET: momentary DPDT, break-before-make: hardware pole J24 COM/NC/NO: isolated second pole -> GPIO6 RESET_REQUEST/GND.
U7 supervises +3V3 (G33 = 3.08 V nominal): watchdog recovery alone CANNOT re-arm.
J21 SAFETY_OK is now a MONITOR OUTPUT. Never drive it externally or bridge to +3V3.

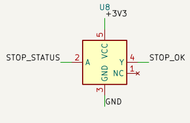






















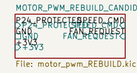




J24: 1=GND/COM, 2=NC_N, 3=NO_N. Never short NC_N to NO_N. Both LOW is forbidden for U2.2.
J25: 1=+3V3, 2=THERMAL_OK, 3=GND, 4=EXT_SAFETY_ENABLE, 3V3 ONLY — NOT A COIL DRIVER.
THERMAL_OK: isolated contact closed to +3V3 when healthy; R22 pulls LOW on open wire.
External STOP and independent thermal limit MUST ALSO directly interrupt the protective contactor coil circuit.
Output buffering, physical fuel disconnection, independent post-purge, layout and bench validation remain pending.



DRAFT_NOT_FOR_ORDER: LM5013 +5V output is intentionally OPEN at root. No connection to DevKit 5V/VIN until USB reverse-feed arbitration is proven on the exact v1.1 board.

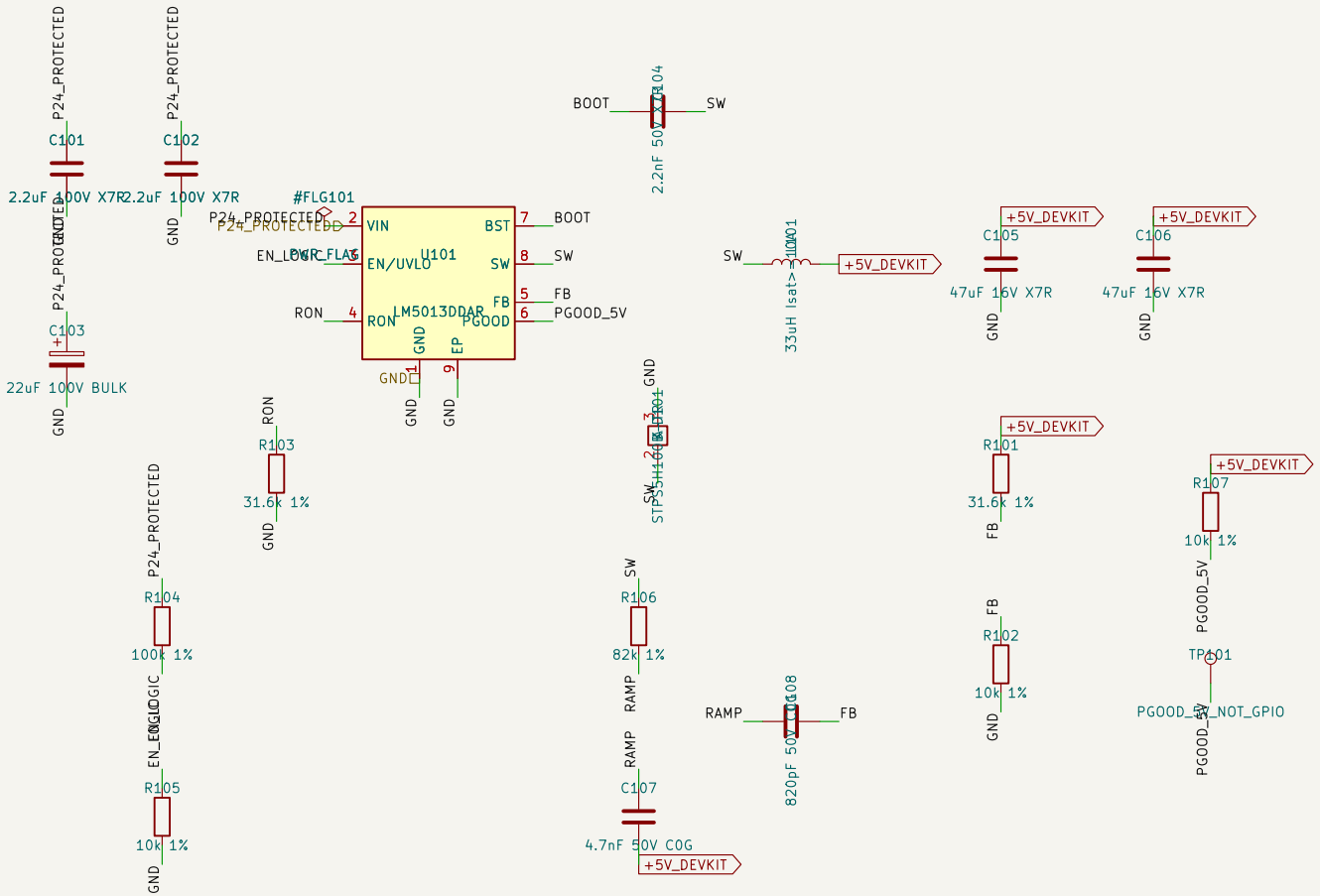


Interface circuit only; not a burner safety approval

Sheet: /
File: Sirokko26201_control_motor_candidate.kicad_sch

Title: Sirokko 262.01 — common controller interfaces

Size: A2	Date: 2026-10-23	Rev: DRAFT-01
KiCad E.D.A. 10.0.7		Id: 1/3



тание логики 24 В → 5 В / 1 А. DRAFT, НЕ ДЛЯ ПРОИЗВОДСТВА.
5013, SNVSCB3 (10.2022), §6-9.2.2.7. Расчётный вход 18-30 В.
о ПОСЛЕ входной защиты; выбросы должны быть ограничены до ≤50 В.
к DevKit и логике на общей PCB. Это не внешний вход 5 В.
онтрольная точка, НЕ GPIO ESP32; не заменяет контроль наличия 24 В.
отключение обратной подпитки USB и силовые каналы здесь ещё не реализованы.
ые места требуют окончательной квалификации; особенно L101 и D101.
щё не выполнена: это отдельный проектный лист, не отдельная внешняя плата.

Sheet: /LOGIC_POWER_24_TO_5V__DRAFT_NOT_FOR_ORDER/		
File: logic_power_24_5_DRAFT.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 2/3

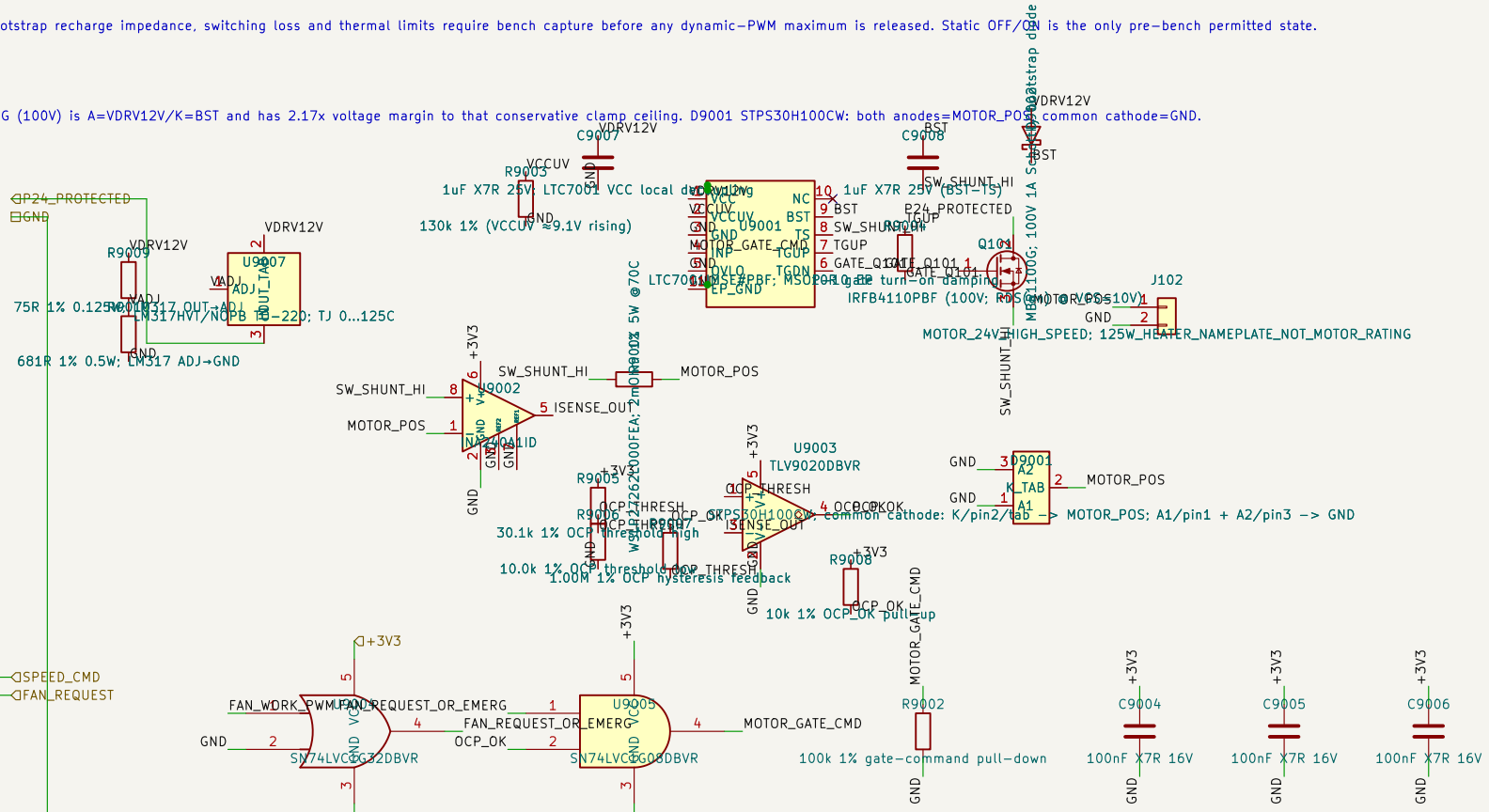
LOGIC CONTRACT — DRAFT, not bench qualified.
INP = ((FAN_CMD AND PWM_SPEED_CMD) OR FAN_EMERG_3V3) AND OCP_OK.
R9002=100k pulls INP LOW when +3V3/gates are off; VIN/VDRV may remain.
U9003 TLV9020DBVR: open-drain OUT4; IN+1, IN−3, V−2, V+5.
Overcurrent pulls OCP_OK LOW and inhibits both PWM and emergency command.
Verify physical shutdown timing and current decay on a safe bench.

CURRENT INTERFACE — DRAFT: 15 A continuous design limit, not measured motor current.
Logic design range +3V3=3.135…3.465 V; 3.0…3.6 V is an additional stress calculation.
Nominal OCP trip at 3.3 V: trip 21.081 A, reset 20.471 A. These are NOT qualified limits.
See OCP_THRESHOLD_REVIEW_RU.md and ocp_threshold_model.json for assumptions.
Coordinate with vehicle fuse, wiring, connector, MOSFET SOA and actual motor startup before release.

ed: A1(1)+A2(3)=GND; K/tab(2)=MOTOR_POS.
ualification stating reverse polarity is NOT qualified.
ermal/SOA and input protection is NOT qualified.

78125Hz; exact driver output−current, bootstrap recharge impedance, switching loss and thermal limits require bench capture before any dynamic−PWM maximum is released. Static OFF/ON is the only pre−bench permitted state.

lification remains OPEN. D9002 MBR1100G (100V) is A=VDRV12V/K=BST and has 2.17x voltage margin to that conservative clamp ceiling. D9001 STPS30H100CW: both anodes=MOTOR_POS, common cathode=GND.



rap; U9001 VCCUV/OVLO/TG wiring; \nFAN_CMD AND PWM gate; OCP comparator threshold/pull−up + latch; TVS/input protection; footprints; validation.\nNo hardware/fire test. 50V transient ceiling is an assumption, not implemented.

Sheet: /MOTOR_PWM_REBUILD_CANDIDATE/
File: motor_pwm_REBUILD.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/3